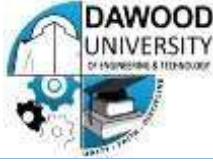

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Experiment 01



CSC-150 – Object Oriented Programming

Name: Prem Kumar

Lab 01

Objective of Experiment: Introduction of Apache NetBeans IDE and Java Programming Environment

Goals: Students will be able to understand the NetBeans IDE and Java Programming Environment..

Name: -----

Required Tools /

Equipment:

- PC
- Java JDK setup <https://www.oracle.com/java/technologies/downloads/>
- Apache NetBeans setup (IDE)
- <https://netbeans.apache.org>

Theory:

Integrated Development Environment (IDE):

Apache NetBeans is a popular and user-friendly Integrated Development Environment (IDE) for Java programming. It allows you to write, compile, run, and debug your code within the same application.

Key Features of NetBeans IDE:

Free and Open Source: NetBeans is free to use.

User-Friendly Interface: Easy for beginners.

Integrated Tools: Editor, build, run, and debugger.

Code Completion: Suggests code as you type.

Debugging Tools: Helps find and fix errors.

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Project Management: Organize multiple files into projects.

Library Support: Add libraries easily for extra features.

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PART 1: Download JDK (Java Development Kit)

1) Download JDK from Oracle/OpenJDK website.

<https://www.oracle.com/java/technologies/downloads/>

The screenshot shows the Oracle Java SE Development Kit 25.0.2 downloads page. The page includes a navigation bar with the Oracle logo and links to Products, Industries, Resources, Customers, Partners, Developers, and Company. Below the navigation bar, there are tabs for Tools and resources, Java downloads (selected), and Java archive. The main content area is titled "Java SE Development Kit 25.0.2 downloads" and includes a note about the JDK 25 binaries being free to use in production and free to redistribute, at no cost, under the Oracle No-Fee Terms and Conditions (NFTC). It also mentions that JDK 25 will receive updates under the NFTC until September 2026, a year after the release of the next LTS. Subsequent JDK 25 updates will be licensed under the Java SE OTN License (OTN) and production use beyond the limited free grants of the OTN license will require a fee. Below this, there are tabs for Linux, macOS, and Windows. A table lists the download links for various operating systems and architectures, including ARM64 Compressed Archive, ARM64 RPM Package, x64 Compressed Archive, x64 Debian Package, and x64 RPM Package. The table columns are Product/file description, File size, and Download. A "Documentation Download" button is located at the bottom left of the page.

Product/file description	File size	Download
ARM64 Compressed Archive	203.36 MB	https://download.oracle.com/java/25/latest/jdk-25_linux-aarch64_bin.tar.gz (sha256)
ARM64 RPM Package	202.96 MB	https://download.oracle.com/java/25/latest/jdk-25_linux-aarch64_bin.rpm (sha256) (JOL 9 GPG Key)
x64 Compressed Archive	203.51 MB	https://download.oracle.com/java/25/latest/jdk-25_linux-x64_bin.tar.gz (sha256)
x64 Debian Package	175.97 MB	https://download.oracle.com/java/25/latest/jdk-25_linux-x64_bin.deb (sha256)
x64 RPM Package	203.07 MB	https://download.oracle.com/java/25/latest/jdk-25_linux-x64_bin.rpm (sha256) (JOL 9 GPG Key)

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2) Install JDK on your PC.

oracle.com/java/technologies/downloads/#jdk25-windows

ORACLE Products Industries Resources Customers Partners Developers Company

Tools and resources Java downloads Java archive

JDK 25 JDK 21

Java SE Development Kit 25.0.2 downloads

JDK 25 binaries are free to use in production and free to redistribute, at no cost, under the [Oracle No-Fee Terms and Conditions \(NFTC\)](#).

JDK 25 will receive updates under the NFTC, until September 2028, a year after the release of the next LTS. Subsequent JDK 25 updates will be licensed under the [Java SE OTN License \(OTN\)](#) and production use beyond the [limited free grants](#) of the OTN license will require a fee.

Linux macOS **Windows**

Product/file description	File size	Download
x64 Compressed Archive	204.59 MB	https://download.oracle.com/java/25/latest/jdk-25_windows-x64_bin.zip (sha256)
x64 Installer	985.27 MB	https://download.oracle.com/java/25/latest/jdk-25_windows-x64_bin.exe (sha256)
x64 MSI Installer	982.04 MB	https://download.oracle.com/java/25/latest/jdk-25_windows-x64_bin.msi (sha256)

Documentation Download

Release Information

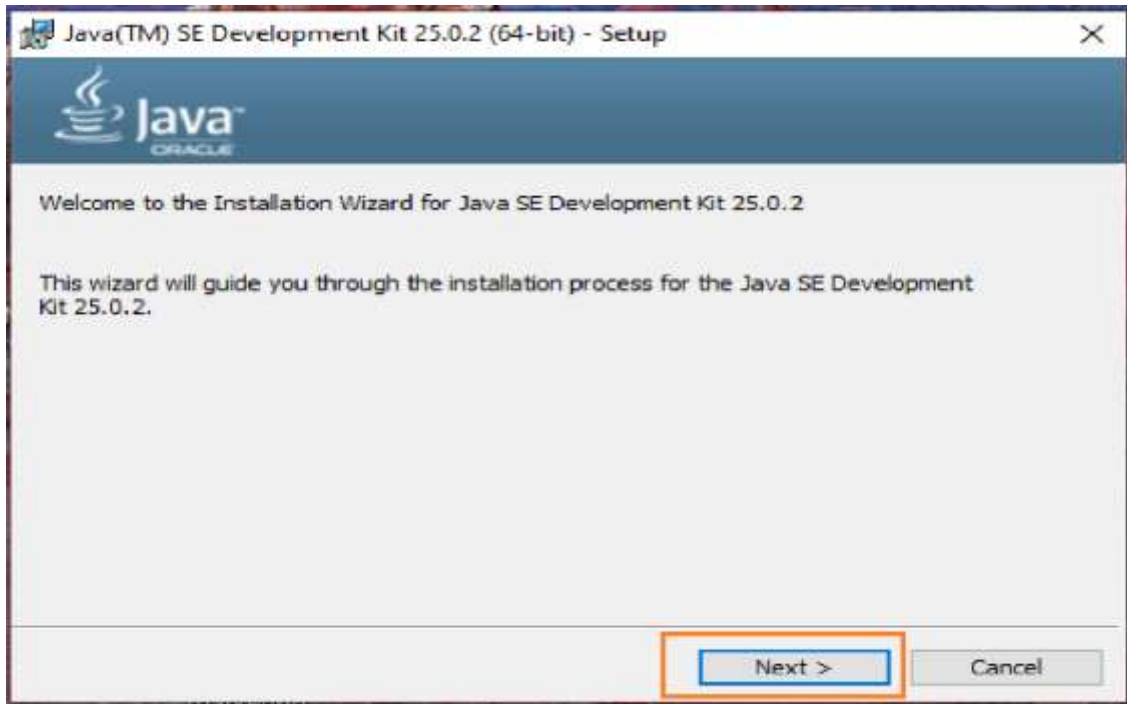
- [Online Documentation](#)
- [Installation Instructions](#)

<https://www.oracle.com/java/technologies/downloads/#jdk25-windows>

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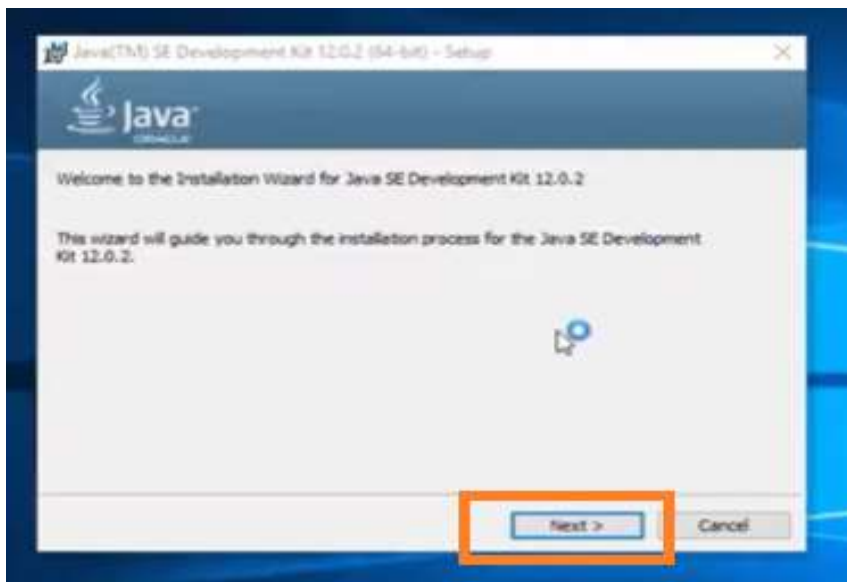
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3) Verify JDK installation in Command Prompt.



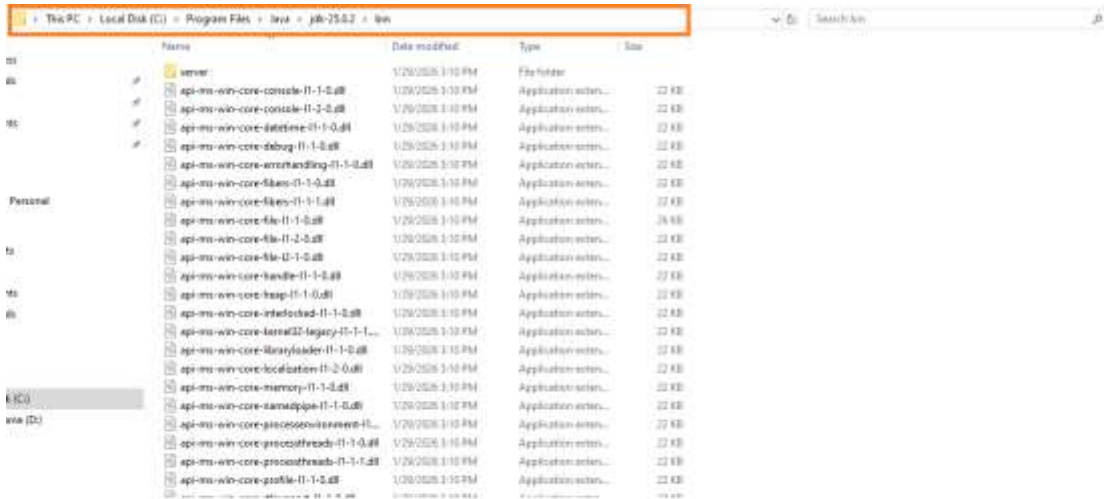
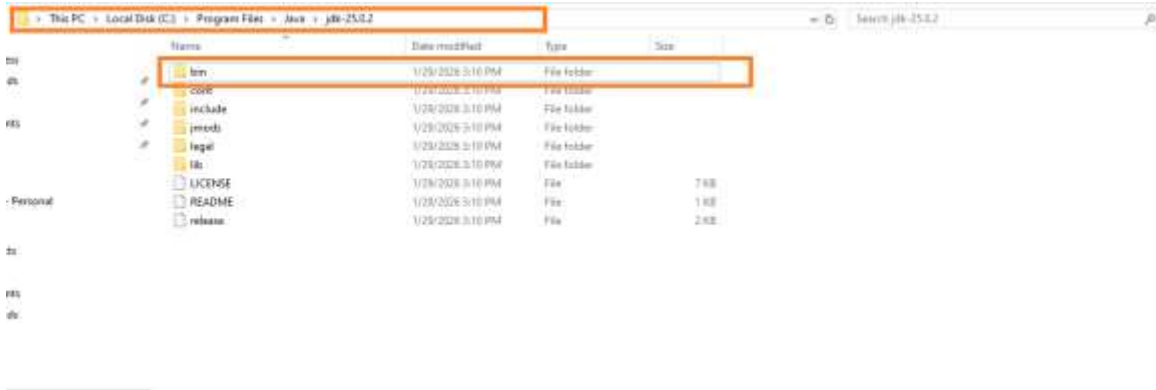
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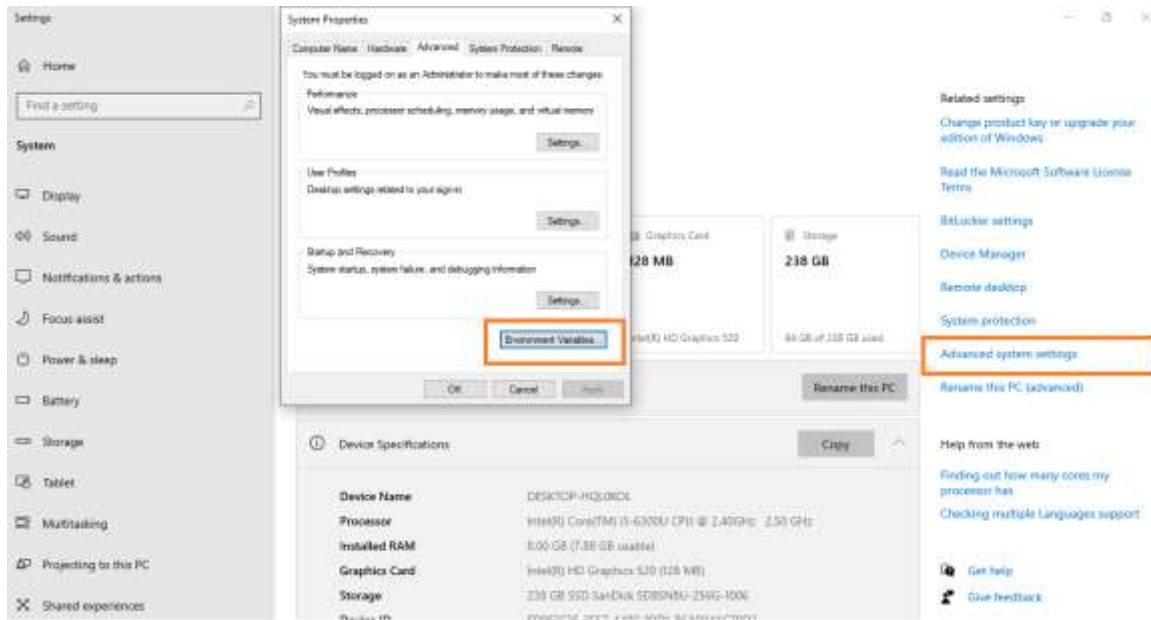
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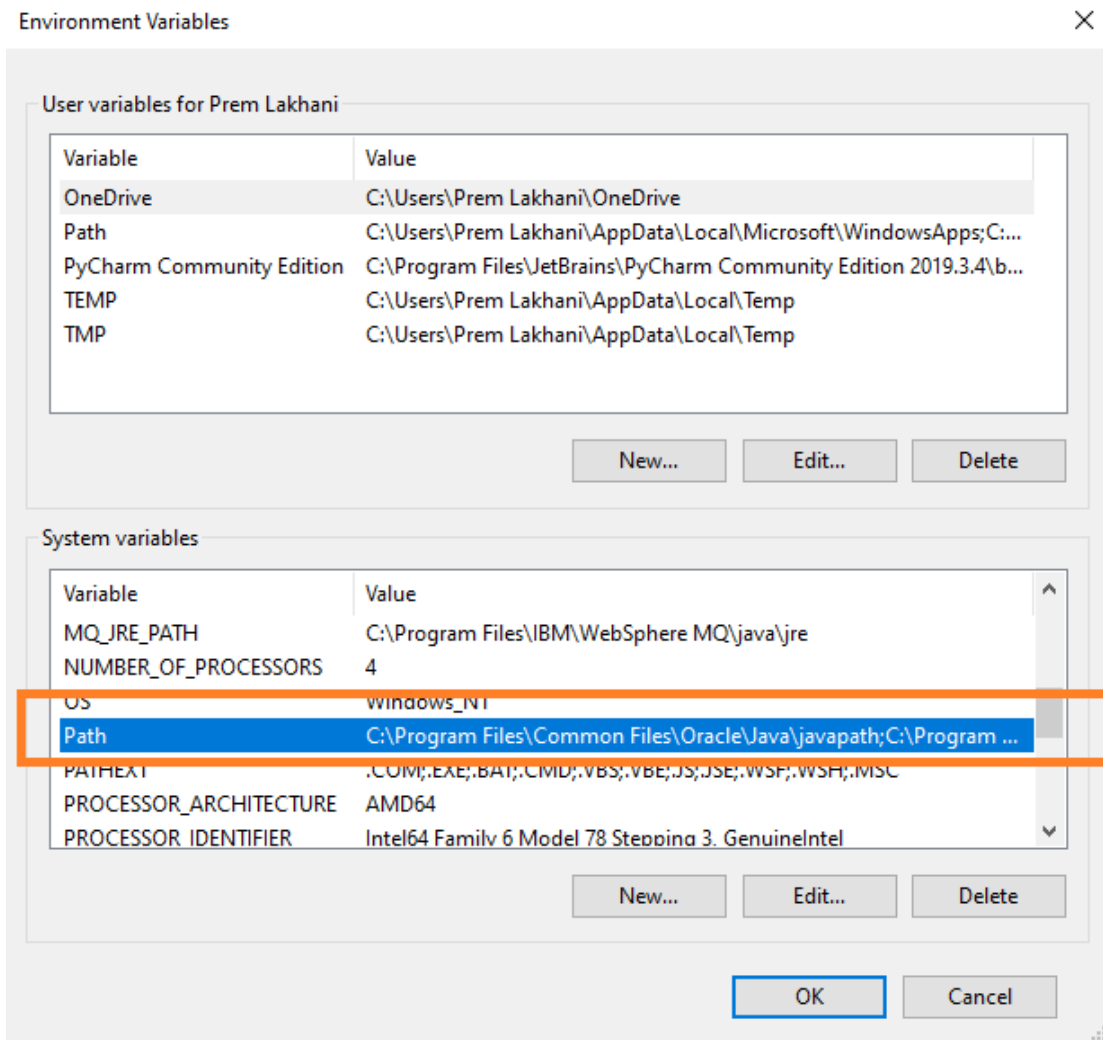
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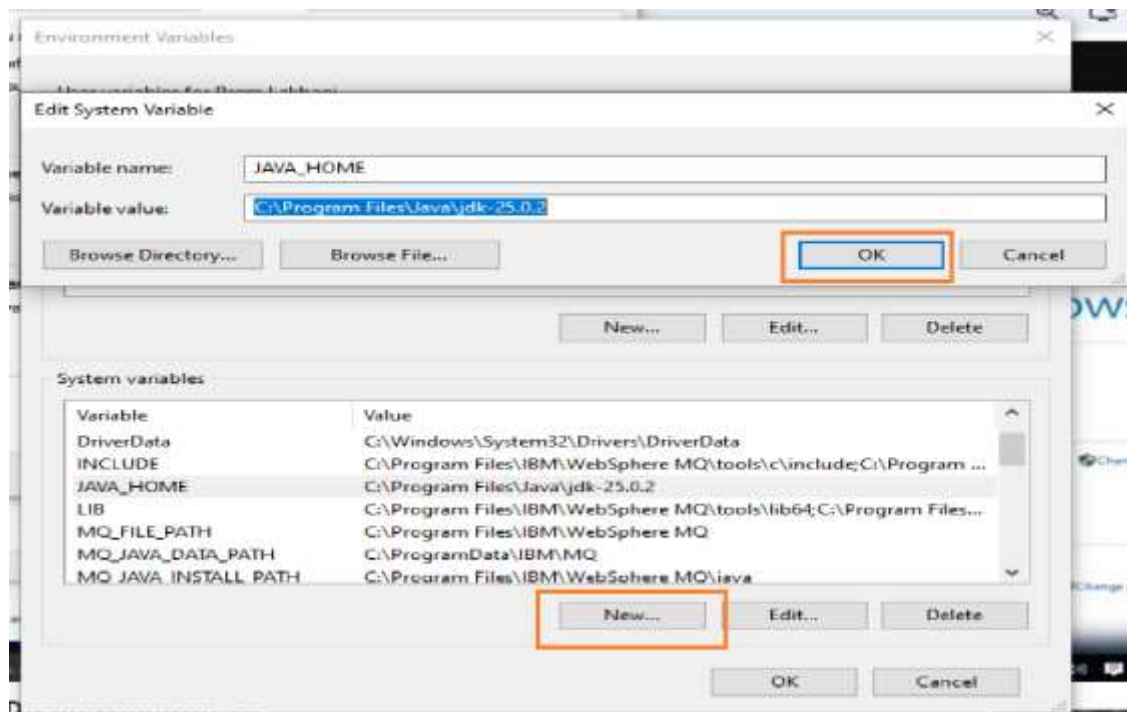
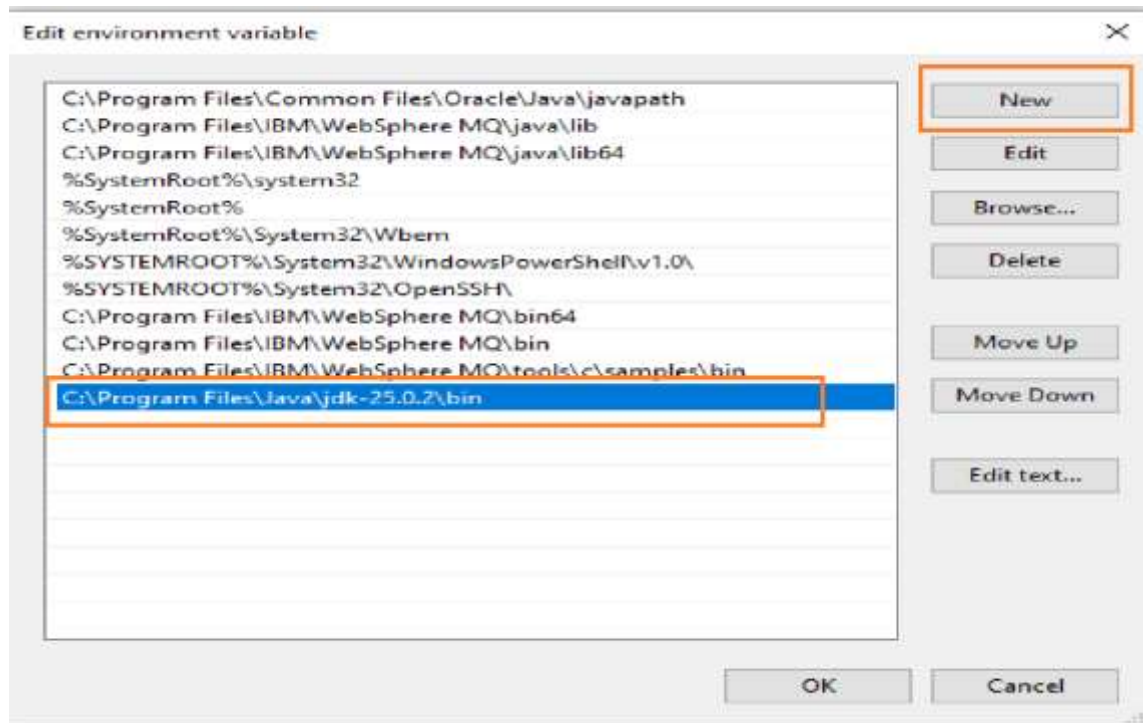
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```
Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Prem Lakhani>java -version
java version "25.0.2" 2026-01-20 LTS
Java(TM) SE Runtime Environment (build 25.0.2+10-LTS-69)
Java HotSpot(TM) 64-Bit Server VM (build 25.0.2+10-LTS-69, mixed mode, sharing)

C:\Users\Prem Lakhani>javac -version
javac 25.0.2

C:\Users\Prem Lakhani>
```

PART 1: Download NetBeans IDE

a- Download NetBeans IDE from: <https://netbeans.apache.org>



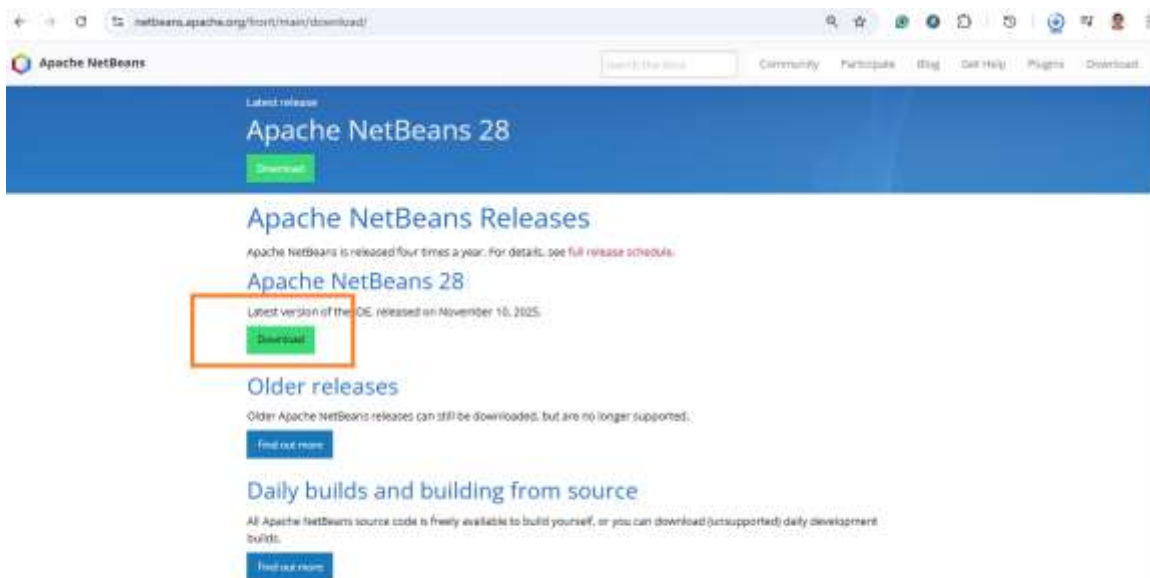
b- Click “Download” for Windows (x64).

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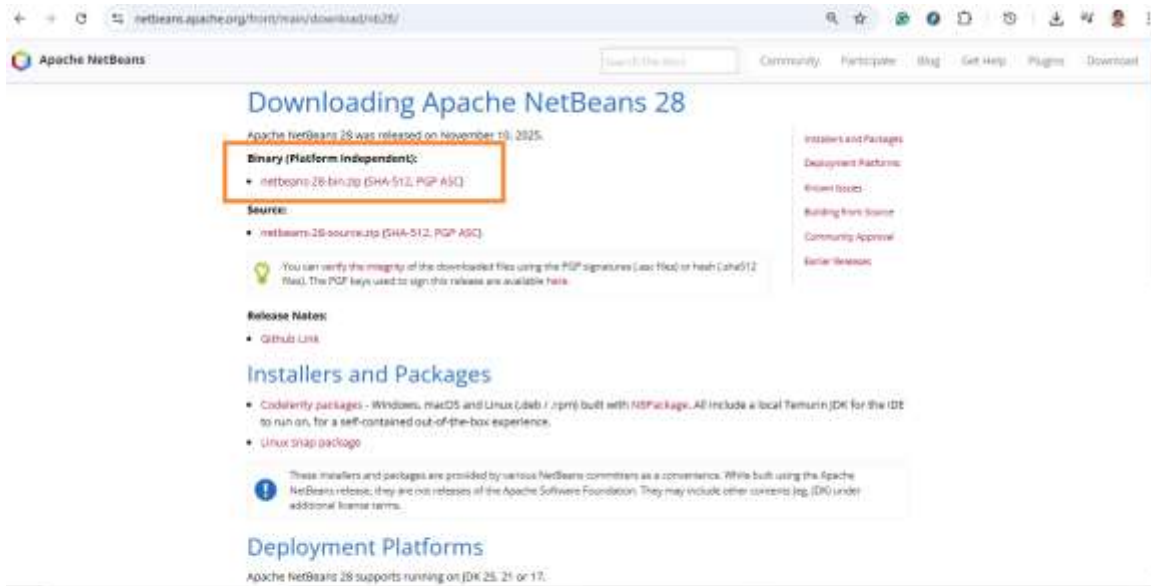


c- Wait while the download completes, then open Downloads folder.



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PART 2: Install NetBeans IDE

1- Run installer (.exe) file.



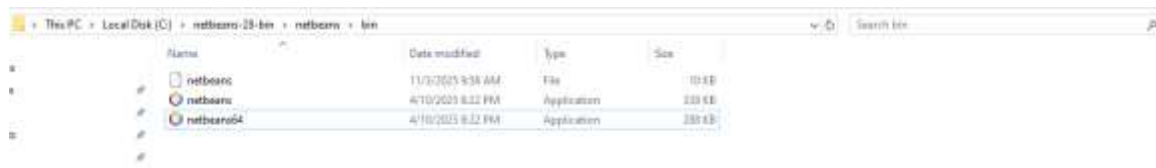
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2- Welcome screen → Click Next.

3- Read and Agree the agreement → Click Next.

4- Choose NetBeans location → Click Next.



5- Confirm Java JDK location → Click Next.

6- Wait while installation is in progress.

7- Done → Click Finish.

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PART 3: Configure NetBeans IDE

1- Open NetBeans from Start Menu.

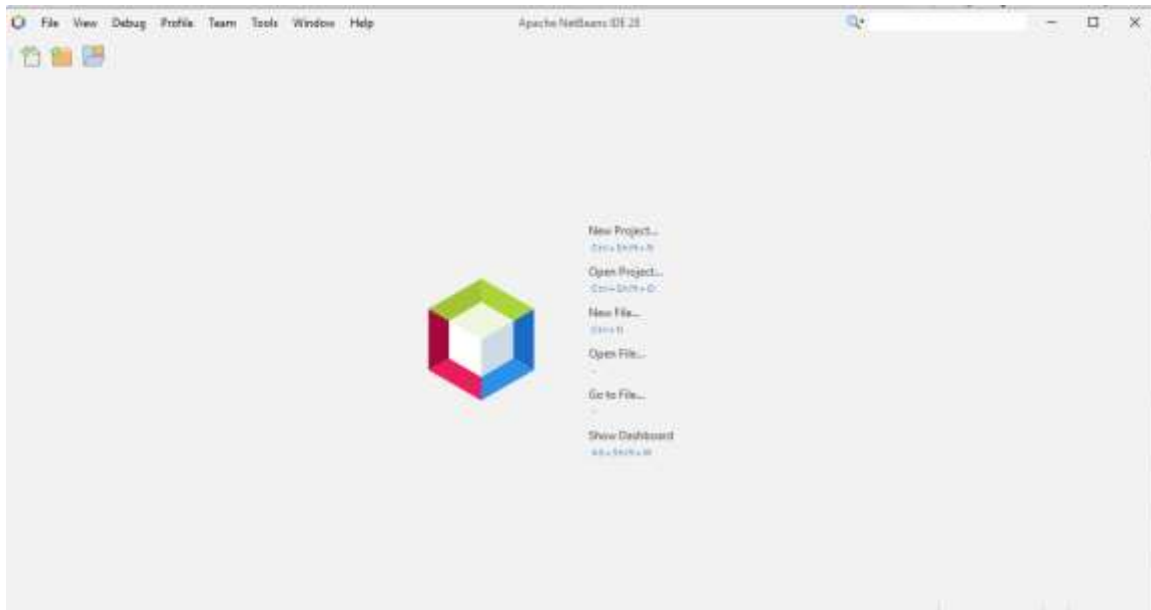
2- If first time configuration prompts appear, keep default settings and continue.

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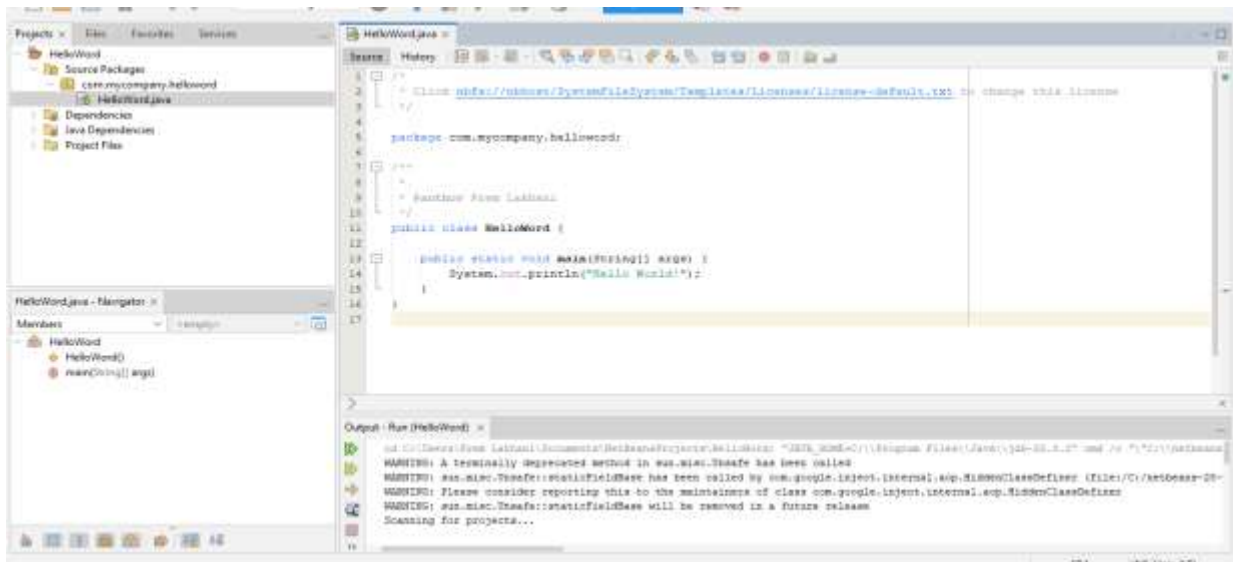
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PART 4: Start with NetBeans

We will write a simple Java program to test NetBeans runs correctly.



1- Select File → New Project.



2- Write your code here (Hello World).

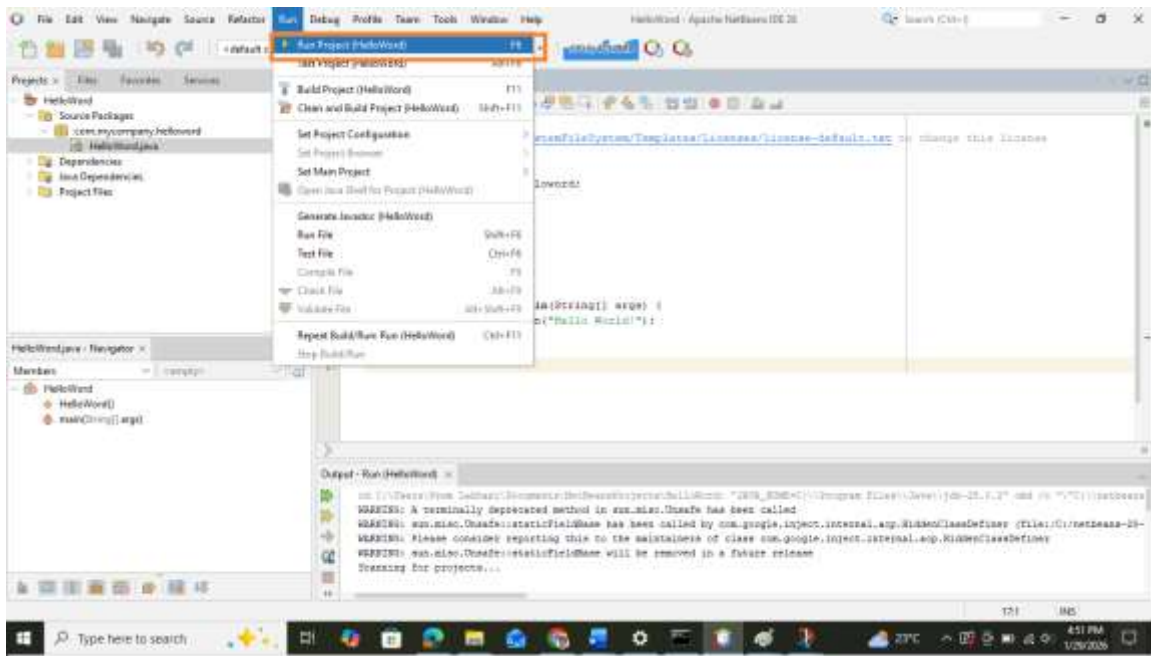
```
class HelloWorld {  
    public static void main(String[] args) {
```


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```
        System.out.println("hello world");  
    }  
}
```

3- Execute → Run Project (or press F6).



10. NetBeans SHORTCUTS KEYS :

You can learn common shortcut keys:

- Run Project: F6
- Build Project: F11
- Save: Ctrl + S
- Find: Ctrl + F
- Comment/Uncomment: Ctrl + /

Student's task:

➤ Install NetBeans on their laptops following the provided installation instructions & take a screenshot as proof of successful setup execution.

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- Create a new Java source file and copy/paste given code & take screenshot of output.

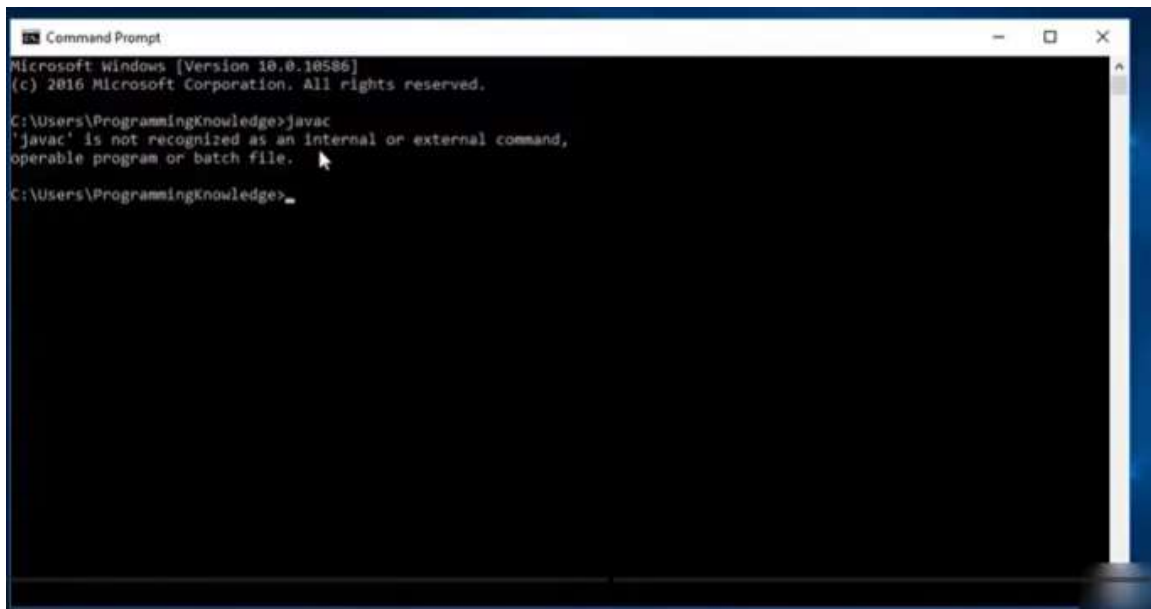
- Make the following changes to the program. What Errors are observed?
 1. Write `System.out.println` as `system.out.println`.
 2. Remove semicolon ; (any one)
 3. Erase any one of brace '{' or '}'.

Date

Course Instructor Sign.

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```
Command Prompt
Microsoft Windows [Version 10.0.18586]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\ProgrammingKnowledge>javac
'javac' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\ProgrammingKnowledge>
```